

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

HardTotals (SR-610/ IM-700)

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Notes

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Contents

Section 1. Introduction	1
Section 2. Details on Settings	2
2.1 Device Information.....	2
2.2 Setting of Devices.....	2
Section 3. Function Details.....	4
3.1 Check Health Method	4
3.2 Property Set Values and Default Values	4
3.3 Supplementary Explanation of Functions	5
3.4 Device Statistics	5
Section 4. Expanded Functions.....	6
4.1 DirectIO Function.....	6
Section 5. Device Specific Programming	9
5.1 Backup of Write Data.....	9
Section 6. Error Information	10
6.1 ResultCode List	10
6.2 Remedial Actions for Principal Errors	13
Section 7. Warnings.....	14

Section 1. Introduction

This manual describes the method of use and related items, including device-specific precautions, when HardTotals are used with EPSON OPOS ADK.

Before the HardTotals can be used, the EPSON OPOS ADK should be installed and the devices to be used should be set using the SetupPOS utility. For setting methods, see the Section 2 of this manual.

This manual applies to the following devices.

Device List

- IM-700 Compact Flash
- SR-610 Compact Flash

The operation of HardTotals is not supported in Windows Vista environment.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the HardTotals.

2.1 Device Information

The DeviceDescription and DeviceName for each model are as follows.

Model Name	DeviceDescription	DeviceName
EPSON IM-700 Compact Flash	EPSON IM-700 Compact Flash HardTotals	IM-700 Compact Flash
EPSON SR-610 Compact Flash	EPSON SR-610 Compact Flash HardTotals	SR-610 Compact Flash

2.2 Setting of Devices

The SetupPOS utility should be used for setting devices. Regarding how to use the SetupPOS utility, refer to the "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

1) Setting CompactFlash Devices

When multiple CompactFlash drives are to be used as HardTotals, it is necessary to perform the "Add device" → "Device specific settings" settings the number of times that corresponds with the number of drives.

2) Device specific Settings

When using a CompactFlash device, it is necessary to specify the drive to which the particular CompactFlash device is allocated.

Open "My Computer" to check which drive CompactFlash is allocated to. It is then necessary to specify the CompactFlash drive in "Device inherent settings".

3) Precautions when Using CompactFlash

- Operation is not guaranteed for other drives than the one CompactFlash is allocated to.
- The default drive is set to "D", so be sure to confirm the CompactFlash drive

and use the SetupPOS utility to correct the setting if necessary.

- After making DeviceEnabled=TRUE, the “\EPSONHARDTOTALS” directory is created in the specified drive. Data is stored in this directory.

This directory cannot be removed even if the OPOS is uninstalled.

- The CompactFlash device can be referenced like a normal physical drive. It is possible to reference it even while EPSON OPOS ADK is in use, but the operation is not guaranteed if files are deleted or copied manually using Explorer, etc.

Section 3. Function Details

This section describes the functions of the HardTotals in details. Supplementary explanation of the parts not described in detail in "OLE for Retail POS Application Programmer's Guide" is also given here.

3.1 Check Health Method

3.1.1 Internal Test

Not supported

3.1.2 External Test

Not supported

3.1.3 Interactive Test

Performs a test of writing and reading of character strings. The input character strings is written to the registered device, and the result read from the device is displayed.

3.2 Property Set Values and Default Values

3.2.1 Capability Set Value

Capability name	Set Value
CapErrorDetection	False
CapSingleFile	False
CapTransactions	False
CapPowerReporting	OPOS_PR_NONE

3.2.2 Property Default Set Values and Setting Ranges

- Free capacity (FreeData)

The free capacity available for the Total devices currently in use. This is also the maximum size when a file is created. This cannot be set.

Setting range $\leq$ TotalsSize

- Capacity of devices (TotalsSize)

The capacity of the Total devices currently in use. This cannot be set.

- Number of files (NumberOfFiles)

The total number of files in the total devices currently in use. This number is increased or decreased by the Create method or Delete method. This cannot be set.

3.3 Supplementary Explanation of Functions

3.3.1 Control of Exclusive Access Rights

Controls the exclusive access rights of the currently used Total devices.

HardTotals supports the method (ClaimDevice, ReleaseDevice) for controlling the exclusive access rights of the entire Total devices, and the methods (ClaimFile, ReleaseFile) for controlling the exclusive access rights in file units.

For details on each method, refer to the "OLE for Retail POS Application Programmer's Guide".

3.4 Device Statistics

The DeviceStatistics function is added in response to the compliance of the "UPOS 1.8".

Please refer to the "EPSON OPOS ADK MANUAL APPLICATION GUIDE Device Statistics" for the details of the Device Statistics.

Section 4. Expanded Functions

This section describes the expanded functions of the HardTotals device.

4.1 DirectIO Function

The usage of the DirectIO method and DirectIOEvent event is described below.

4.1.1 DirectIO Method

Syntax **DirectIO** *Command* As Long, *pData* As Long, *pString* As String

Parameter	Explanation
<i>Command</i>	Command number
<i>pData</i>	Sets the specific command to ON/OFF (0=OFF/ values other than 0 =ON)
<i>pString</i>	Not used

Remarks The particular processing is performed in accordance with the command number.

Command number =TOT_DI_BACKUP

If a value other than 0 is entered for *pData* and the command is executed, since then, a backup file containing the write data is created in the local hard disk before actual writing to the Total device (file) is performed in the methods (Create, Delete, Write, SetAll, Rename) for writing to Total device.

Creation of the backup file means that if system failure occurs during writing of data to Total device, the Total device can be restored next time the application is started up.

Entering 0 for *pData* and executing cancels creation of a backup file.

This setting changes the settings of all HardTotals opened as a Share.

Return Either of the following values is returned, and placed in the ResultCode property.

Value	Meaning
OPOS_SUCCESS	DirectIO was successful.
OPOS_E_CLOSED	Control is closed.
OPOS_E_CLAIMED	Another object has issued ClaimDevice.
OPOS_E_ILLEGAL	The command is illegal or not supported.
	The exclusive access right cannot be achieved.
Others	See the "ResultCode List".

Prerequisites Open

4.1.2 DirectIOEvent Event

Syntax **DirectIOEvent** *EventNumber* As Long, *pData* As Long, *pString* As String

Parameter	Explanation
<i>EventNumber</i>	Event number TOT_DIE_FORWARD: Condition after execution of method was restored. TOT_DIE_ROLLBACK: Condition before execution of method was restored.
<i>pData</i>	File number
<i>pString</i>	Not used.

- Remarks** When a backup file is created by the DirectIO method's backup function, the restored condition of the Total device (file) established by the backup file is communicated to the application if system failure occurs during writing to a Total device, and there is a risk that the Total device contents may have been corrupted. The file numbers should be read as follows.
- 1: Means that a failure has occurred during execution of the Create, Delete, Rename methods for restoration of file information.
 - 2: Means that a failure has occurred during rebuilding of a device file using the DirectIO method for restoration of the entire device.
 - 1-9: Means that a failure has occurred during execution of the Create, Write, and SetAll methods for restoration of the file data held in the corresponding file number. The file number corresponds to the order in which the file name was established when the FindByIndex method was use to obtain the file name.

Section 5. Device Specific Programming

This section describes the programming specific to the HardTotals device.
The coding examples are real examples created in Microsoft Visual Basic.

5.1 Backup of Write Data

The DirectIO method can be used to specify whether a backup file should be created on the local hard disk when the method for writing to a Total device is executed.

For details on the DirectIO method, see the Section 4 of this manual.

When a backup file should be created:

```

Dim RC As Long
Dim Data As Long
Dim Dummy As String

Data = TRUE      ' Specify FALSE when file should be not created
RC = OPOSTotals1.DirectIO (TOT_DI_BACKUP, Data, Dummy)
If RC = OPOS_SUCCESS Then
    'Start Backup
Else
    'Error
End If

```

Section 6. Error Information

This section describes about the error information when executing properties of the methods of the HardTotals. The common properties and methods are informed in "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE GENERAL DEVELOPMENT". Refer to the guide for more information.

6.1 ResultCode List

The ResultCode and ResultCodeExtended codes when methods are executed are as follows.

Method Name	ResultCode	ResultCodeExtended	Meaning
ClaimFile	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Handle is illegal.
		OPOS_EX_BADPARAM + 2	Timeout period is illegal.
		OPOS_ETOT_DELETE	Total file is not possible to delete.
	OPOS_E_TIMEOUT	0	Refer to <i>UPOS Specifications</i>
Create	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_EXISTS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_EXTENDED	OPOS_ETOT_NOROOM	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	File name is illegal.
		OPOS_ETOT_READFILE	Error during reading from device.
		OPOS_ETOT_WRITEFILE	Error during writing to device.
		OPOS_ETOT_INIT_DEVICE	Device is not ready.
	OPOS_E_FAILURE	OPOS_EX_UNAUTHORIZED	There was insufficient permission to access for processing.
Delete	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_NOEXIST	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	OPOS_ETOT_LOCKED	Method under execution by same object.
	OPOS_E_FAILURE	OPOS_EX_UNAUTHORIZED	There was insufficient permission to access for processing.

Method Name	ResultCode	ResultCodeExtended	Meaning
Find	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_NOEXIST	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
FindByIndex	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
Read	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
		OPOS_ETOT_LOCKED	Method under execution by same object.
		OPOS_ETOT_READFILE	Error during reading from device.
		OPOS_EX_BADPARAM+3	The 3 th parameter is illegal.
		OPOS_EX_BADPARAM+4	The 4 th parameter is illegal.
ReleaseFile	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
		OPOS_EX_NOTCLAIMED	Not claimed.
Rename	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_EXISTS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	File name is illegal.
		OPOS_ETOT_LOCKED	Method under execution by same object.
		OPOS_ETOT_WRITEFILE	Error during writing to device.
	OPOS_E_FAILURE	OPOS_EX_UNAUTHORIZED	There was insufficient permission to access for processing.

Method Name	ResultCode	ResultCodeExtended	Meaning
SetAll	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
		OPOS_ETOT_LOCKED	Method under execution by same object.
		OPOS_ETOT_READFILE	Error during reading from device.
		OPOS_ETOT_WRITEFILE	Error during writing to device.
	OPOS_E_FAILURE	OPOS_EX_UNAUTHORIZED	There was insufficient permission to access for processing.
Write	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
		OPOS_EX_BADPARAM + 2	Write data does not comply with BinaryConversion. Refer to <i>UPOS Specifications</i>
		OPOS_ETOT_LOCKED	Method under execution by same object.
		OPOS_ETOT_READFILE	Error during reading from device.
		OPOS_ETOT_WRITEFILE	Error during writing to device.
		OPOS_EX_BADPARAM+3	The 3 th parameter is illegal.
		OPOS_EX_BADPARAM+4	The 4 th parameter is illegal.
	OPOS_E_FAILURE	OPOS_EX_UNAUTHORIZED	There was insufficient permission to access for processing.
ValidateData	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
RecalculateValidationData	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>

Method Name	ResultCode	ResultCodeExtended	Meaning
BeginTrans	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
CommitTrans	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>
Rollback	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_CLAIMED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i>
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i>

6.2 Remedial Actions for Principal Errors

- Timeout error occurs during claim for exclusive access rights. Likewise, none of the methods can be executed.
 - There is a possibility that another application already has obtained exclusive access rights to the specified device (file). Wait until the exclusive access right is released, or close the particular application, and then execute again.
- When creating a file or changing the file name, OPOS_E_ILLEGAL is returned.
 - This is occurs when the file name is improper. The file name may consist of up to 10 characters (10 bytes), and character codes smaller than 0x20 (space) cannot be used.

Section 7. Warnings

- When using the DirectIO method to issue a command, be sure to read the instructions on the particular command in the product manual.
- Do not connect or disconnect a CompactFlash device while the power is ON. There is a risk that the system may be destroyed.
- Total file with file name is supported. However, the file name is restricted to 10 one-byte characters (5 two-byte characters).
- An advanced error detection function is not supported.
- The Transaction function is not supported. Consequently, the BeginTrans, CommitTrans, RecalculateValidationData, Rollback, and ValidateData methods are not supported.
- The FreeData property returns the actual file size that can be created with the Create method.
- During actual execution of a method, a method cannot be executed on the same object as a rule.
- The CheckHealth, DirectIO, Create, Delete, and Rename methods cannot be executed while a method is being executed on another object.
- During execution of a method (Read, Write, SetAll) for reading or writing to file, temporary exclusive access right to the file is obtained regardless of whether or not exclusive access right to the file has been obtained.
- During execution of CheckHealth and methods (DirectIO, Create, Delete, Rename) for rewriting device file information, temporary exclusive access right to the file is obtained regardless of whether or not exclusive access right to the file has been obtained.